

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 7: Due by 11:59pm on Tuesday April 2, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [10 points] Prove the inequality $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1 \leq n\|\mathbf{x}\|_\infty$, which establishes an equivalence of norms. Here, n is length of $\mathbf{x}$ so that $\mathbf{x} \in \mathbb{R}^n$.

2. [20 points] Let $A = \begin{bmatrix} .99 & .98 \\ 1 & .99 \end{bmatrix}$.

(A). [10 points] Use the Gaussian elimination (GE) code from class to solve for $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ with $\mathbf{b} = \begin{bmatrix} 0.101 \\ 0.100 \end{bmatrix}$.

Then solve $A\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ for $\tilde{\mathbf{x}}$, where $\tilde{\mathbf{b}} = \mathbf{b} + \delta\mathbf{b}$ and $\delta\mathbf{b} = \begin{bmatrix} 0.001 \\ 0.000 \end{bmatrix}$. Comment on what you observe.

(B). [10 points] Calculate the condition number $\text{cond}(A) = \|A\|_2 \|A^{-1}\|_2$ and obtain upper and lower bounds for the relative difference $\|\delta\mathbf{x}\|_2 / \|\mathbf{x}\|_2$, where $\delta\mathbf{x} = \mathbf{x} - \tilde{\mathbf{x}}$. How does this relate to your results to Problem 2A.

3. [30 points] Consider the system of differential equations $\frac{d}{dt}\mathbf{y} = (\gamma A - \beta I)\mathbf{y}$, where $A \in \mathbb{R}^{1000 \times 1000}$ is the random matrix from class in which each entry A_{ij} is either a 1 with probability $p = 0.005$ or a 0 with probability 0.995 and I is a size-1000 identity matrix. This equation models epidemic spreading defined over a network of face-to-face interactions (encoded by A), and parameter $\gamma > 0$ is an *infection rate* and $\beta > 0$ is a *healing rate*. Each $y_j^{(t)}$ gives the probability that person j is infected at time t . We will assume $\gamma = 0.14$, $\beta = 0.1$, and an initial condition in which $y_1^{(0)} = 0.1$ and $y_j^{(0)} = 0$ for $j > 1$.

(A). [10 points] Solve the system of equations using the forward Euler (FE) method $\mathbf{y}^{(i+1)} = \mathbf{y}^{(i)} + hM\mathbf{y}^{(i)}$, where $M = (\gamma A - \beta I)$, $t \in [0, 10]$, $N = 500$, and $h = 1/50$. Plot the approximated trajectories $y_j^{(t)}$ for $t \in [0, 10]$ the first 10 people, $j = 1, \dots, 10$.

(B). [10 points] Repeat Problem 3A, but now solve the IVP using the backward Euler method (BE), which requires solving a linear equation at each step: $(I - hM)\mathbf{y}^{(i+1)} = \mathbf{y}^{(i)}$. Use the residual-correction method to solve the linear system with a tolerance of 10^{-15} (i.e., $\|r\| < 10^{-15}$) and $C = I$ as the approximate inverse.

(C). [10 points] Repeat Problem 3B, but implement your residual-correction method using sparse matrices. Compare the runtimes for with and without using sparse matrices.

4. [40 points] Consider the boundary value problem (BVP) $\frac{d^2 y}{dt^2} = 4y$ with boundary conditions $y(0) = 1$ and $y(1) = 3$, which has the true solution $y(t) = c_1 e^{2t} + c_2 e^{-2t}$ with $c_1 = \frac{3-e^{-2}}{e^2-e^{-2}}$ and $c_2 = 1 - c_1$. The finite difference approximation to the BVP with time mesh $t_i = 1/N$ for $i = 0, \dots, N$ can be by setting $y_0 = 1$, $y_N = 3$ and the remaining y_i values are the solution $\mathbf{y}$ to the linear equation $M\mathbf{y} = \mathbf{b}$, where M is an $(N-1) \times (N-1)$ square matrix and $\mathbf{b}$ is a length- $(N-1)$ vector and they take the form:

$$\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ \vdots \\ 0 \\ -3 \end{bmatrix}, \quad M = \begin{bmatrix} -2 - 4/N^2 & 1 & 0 & \dots & 0 \\ 1 & -2 - 4/N^2 & 1 & \ddots & \vdots \\ 0 & 1 & -2 - 4/N^2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \dots & 0 & 1 & -2 - 4/N^2 \end{bmatrix}.$$

- (A). [10 points] Let $N = 4$ and use the Gauss-Jacobi method with a tolerance of 10^{-15} to solve for $\mathbf{y}$. (Use the matrix formulation of GJ given in class.) Make a plot comparing the estimates y_i at t_i to the true solution $y(t)$. Plot the errors $|y_i - y(t_i)|$ at times t_i in a second plot.
- (B). [10 points] Repeat Problem 4A, but use the Gauss-Seidel method instead.
- (C). [10 points] Repeat Problem 4A, but convert M into a sparse matrix format using `'csr_matrix()'` then use SciPy's built-in sparse linear solver `'scipy.linalg.sparse.spsolve()'`.
- (D). [10 points] Record the runtimes for the GJ and GS methods, as well as for the sparse linear solver, and study how their runtimes change as one increases $N = 4, 40$, and 400 .